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Evaluating Temporal Dimensionality Reduction Methods for ERP-Structured EEG Data

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About Me

2017-2021

Bachelor’s of Engineering in Computer Science (India)

2021-2023

2 years of work experience as System Developer in Cybersecurity (India)

2023-Present

Currently pursuing a Master’s in Computer Science

Major: Service technology and Engineering

Motivation

The **complexity** is the fundamental reason why advanced dimensionality reduction techniques are necessary.

High dimensionality
Many channels
Many subjects

Noise
Consists of artifacts

Feature Selection
Various features to focus on (time/frequency-domain features)

Dimensionality reduction helps in interpreting EEG visualization.

Why are PCA or t-SNE algorithms not enough for EEG data analysis?

Linear Limitations (PCA):

Principal Component Analysis identifies maximal variance but assumes a flat Euclidean geometry, often failing to capture the "curvature" of brain state transitions

PCA, t-SNE, and UMAP are "time-agnostic":

- They treat each time point as an independent, isolated observation.
- They ignore the temporal nature of EEG signals, where the neural state at ‘t’ is naturally dependent on the state at ‘t-1.’
- Groups points by similarity.

Advanced Time-preserving methods

Non-Linear Advantage:

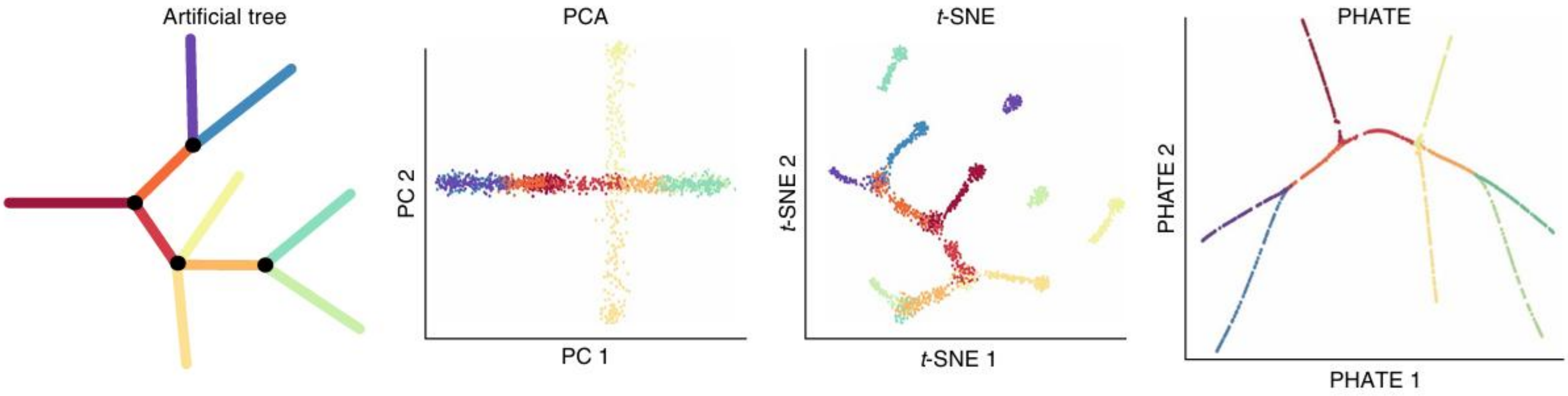
Methods like PHATE and T-PHATE preserve both local and global structures

Highlight continuous progressions and ERP components in different conditions.

TPHATE specifically focuses on the time autocorrelation nature of the data: it groups points by **similarity + sequence.**

Shows the evolution from P100 towards P300.

PHATE: Potential of heat diffusion for affinity-based transition embedding



Core idea:

1. Calculate similarity between local data points Using alpha decay kernel

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2. Calculate distances between data points globally: Diffusion maps

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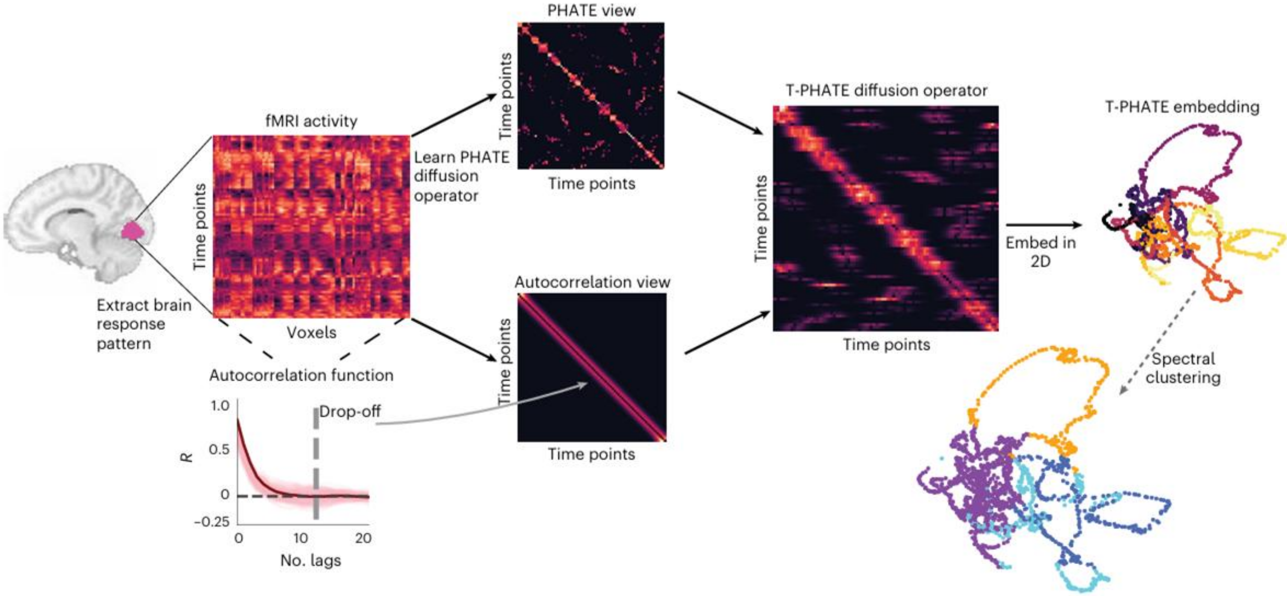
3. Map potential distance information into low dimensions Using MDS

TPHATE: Temporal PHATE



Temporal PHATE to analyze the sequential order of data

Dual-Diffusion Operator: Geometric similarity (PHATE) + time autocorrelation

Busch et al., 2023)

Research Questions & Hypotheses

1. Does 'time-aware' learning (T-PHATE) outperform 'time-agnostic' learning (PCA/t-SNE/UMAP) for ERP-related EEG data?
2. Which metrics best evaluate temporal preservation in dimensionality reduction?

Hypotheses: T-PHATE will *significantly* outperform standard methods in:

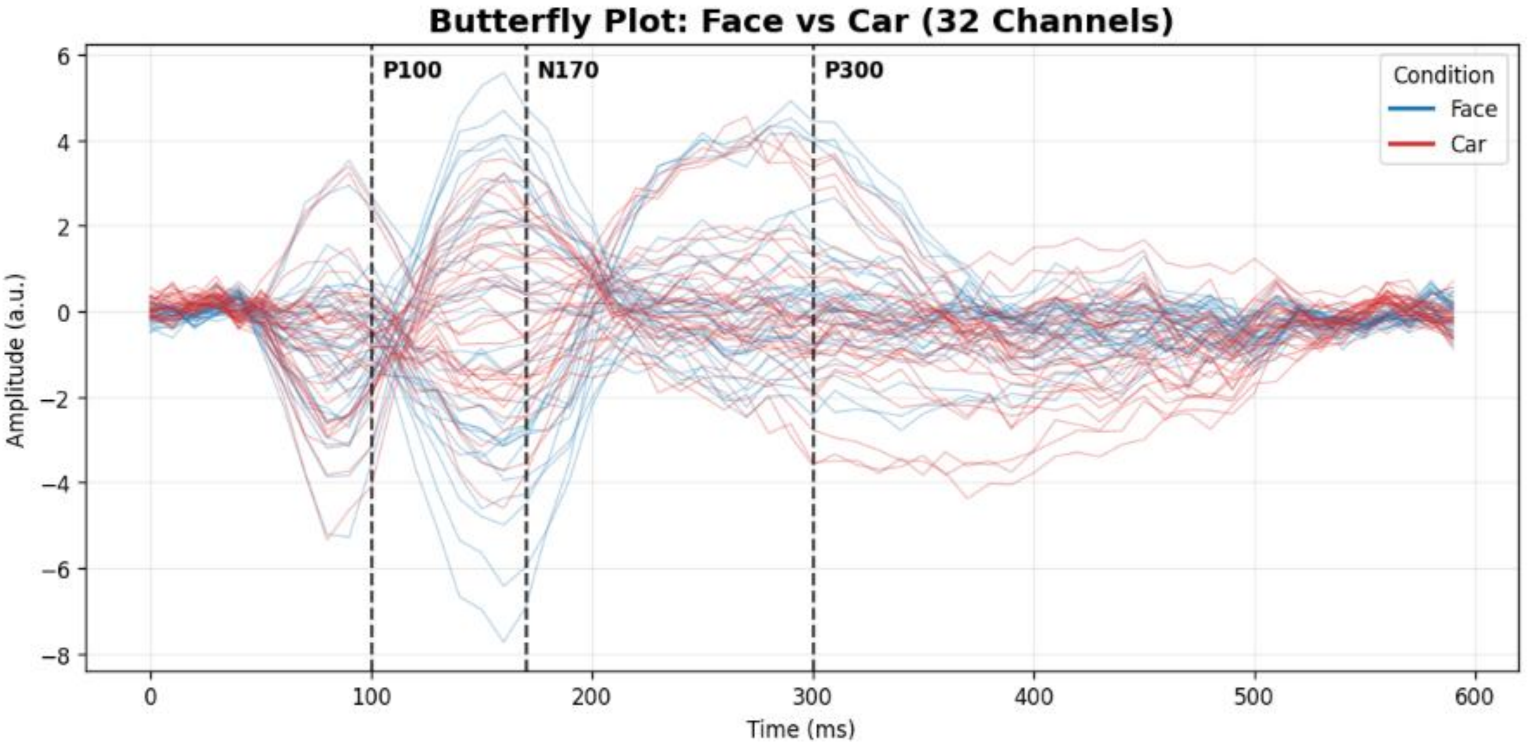
- Preserving the ground-truth structure of EEG/ERP data
- Show the continuous ERP component trajectory from P100-N170-P300 even in multiple conditions

Methods and Approach

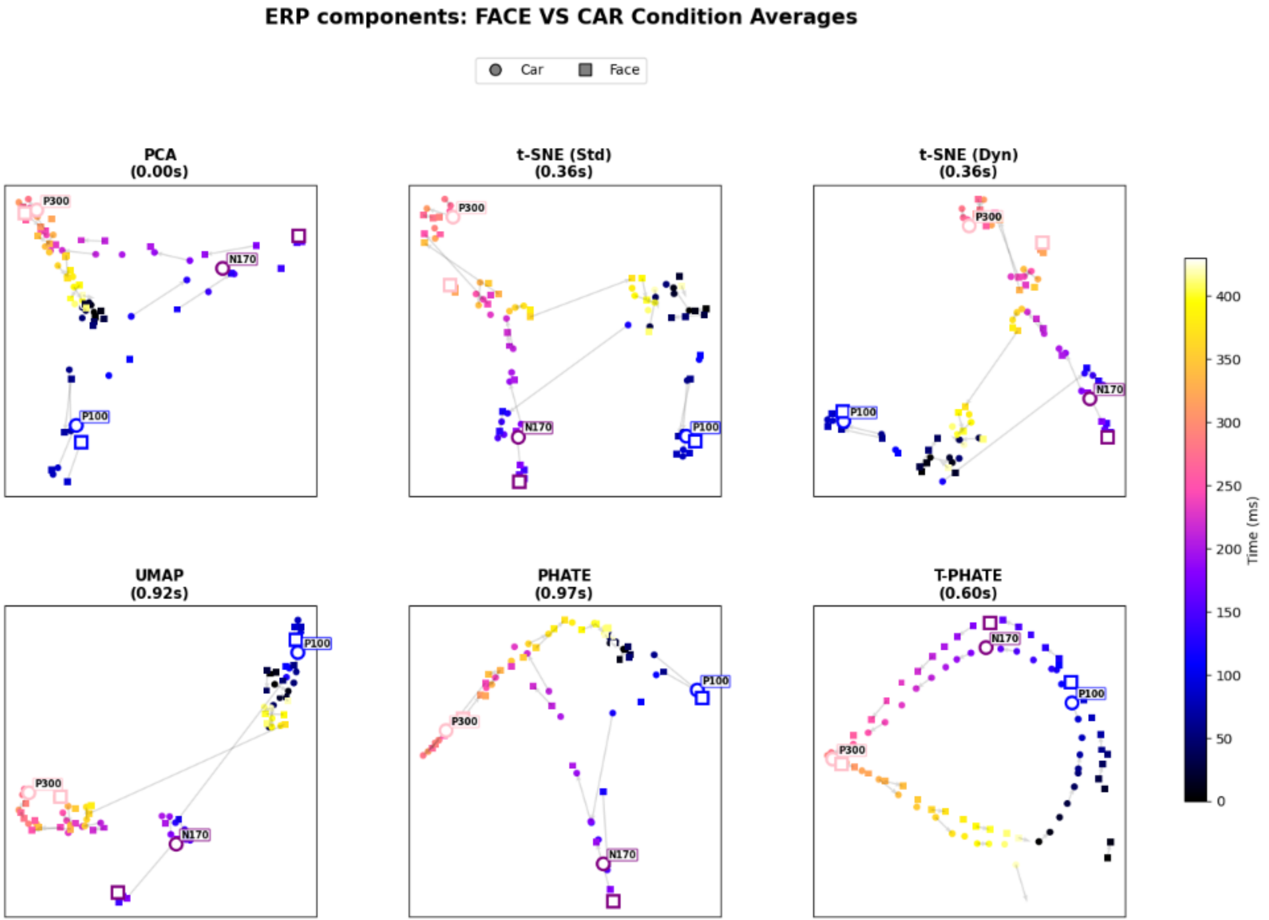
Using **Julia** and **Python**

1. High-speed generation of EEG data using UnfoldSim.jl
   a. Julia enables the simulation of complex multi-subject designs in seconds.
   b. Customize noise at single-subject and multi-subject levels (essential for TPHATE and PHATE as they denoise the data as well)
2. Apply various dimensionality reduction techniques in Python, like
   a. PCA
   b. T-SNE
   c. UMAP
   d. PHATE
   e. TPHATE

Data Simulation: Single-Subject Design



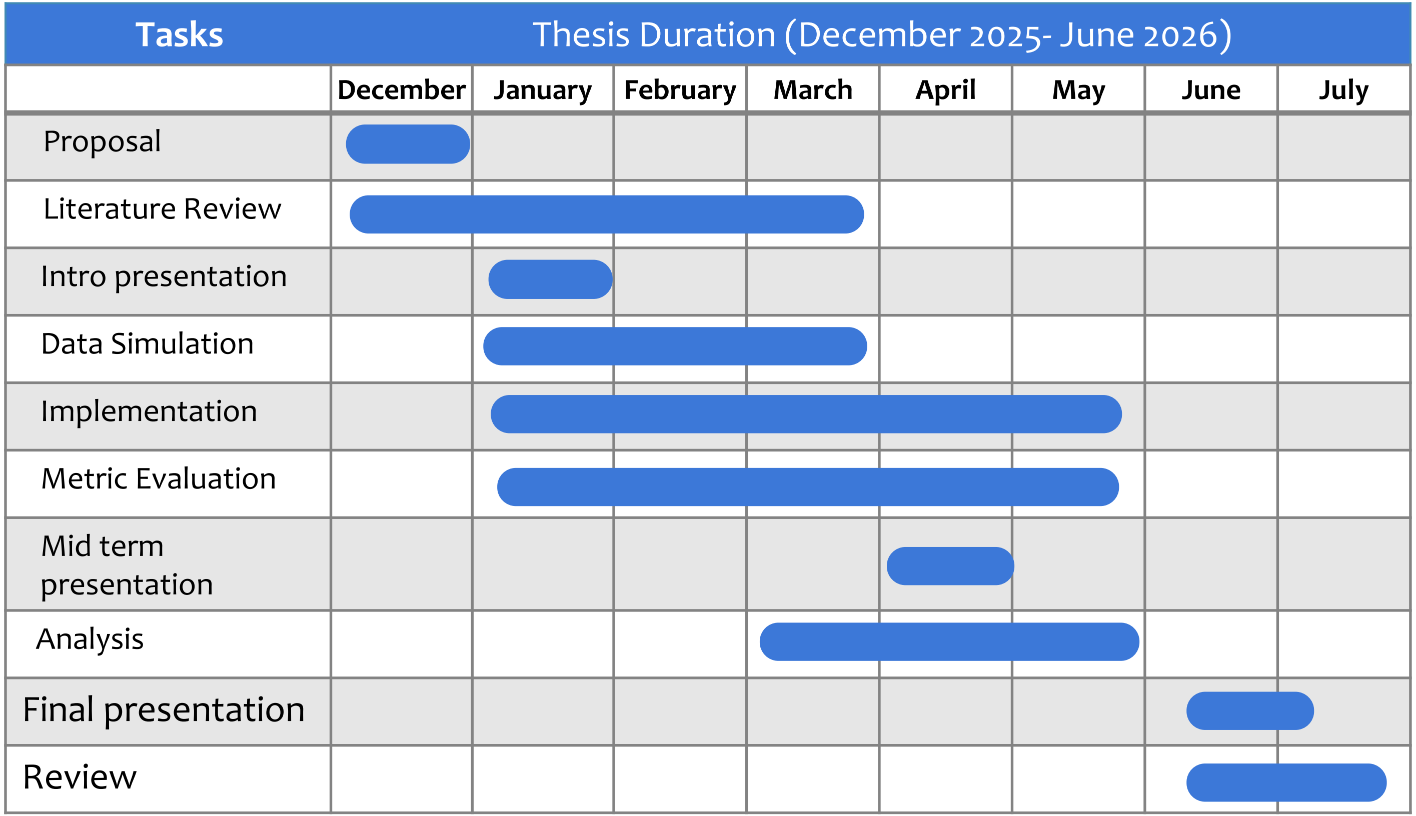
Dimensionality reduction methods on simulated data



Algorithms

	PCA	t-SNE	UMAP	PHATE	T-PHATE
Geometry	Linear	Non-Linear	Non-Linear	Non-Linear	Non-Linear
Structure Preservation	Maximum variance, global structure	Local structure	Local & Global structure	Local & Global structure	Local & Global structure
Temporal Data Handling	No	No	No	Good for trajectories, but does not explicitly use time	Explicitly incorporates time.

Timeline


Tasks
Thesis Duration (December 2025- June 2026)
December
January
February
March
April
May
June
July
Proposal
Literature Review
Intro presentation
Data Simulation
Implementation
Metric Evaluation
Mid term presentation
Analysis
Final presentation
Review

References

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- Moon, K. R., van Dijk, D., Wang, Z., *et al.* (2019). Visualizing structure and transitions in high-dimensional biological data. *Nature Biotechnology, 37*(12), 1482–1492. https://doi.org/10.1038/s41587-019-0336-3
- Busch, E. L., Huang, J., Benz, A., *et al.* (2023). Multi-view manifold learning of human brain-state trajectories. *Nature Computational Science, 3*(3), 240–253. https://doi.org/10.1038/s43588-023-00419-0
- Zhang, J., Yang, Y., Liu, T., Shi, Z., Pei, G., Wang, L., Wu, J., Funahashi, S., Suo, D., Wang, C., & Yan, T. (2023). Functional connectivity in people at clinical and familial high risk for schizophrenia. *Psychiatry Research, 328*, 115464.